

CORE 2A

Motion in 2D - Projectile Motion

Supported familiar application: five learner-eligible authored practice twins.

Use pattern

Attempt each problem first. Then compare your route with REPRESENT -> DECIDE/TRANSFORM -> VERIFY. These are authored practice items, not source questions.

Practice 1

A particle has $x(t)=10t$ and $y(t)=18t-2t^2$ in metres. Find the direction of its instantaneous velocity at $t=2$ s.

Your independent attempt

Crux move

Represent velocity by derivatives of position.

Complete reasoning route

1. REPRESENT - Write $v_x = dx/dt$ and $v_y = dy/dt$.
2. TRANSFORM - Differentiate and evaluate both at $t=2$ s.
3. CONNECT - Use the signed velocity-component ratio for direction.
4. VERIFY - Confirm the velocity arrow is tangent to the trajectory.

Answer

$v_x=10$ m/s, $v_y=10$ m/s, so the direction is 45 degrees above +x.

Independent check

Do not use y/x ; that is a position-vector direction, not the tangent velocity.

Practice 2

A projectile launched at 45 degrees reaches 80 m maximum height. On returning to the ground it retains 60% of its kinetic energy and rebounds at 30 degrees above horizontal. Find its new maximum height.

Your independent attempt

Crux move

Convert retained kinetic energy to retained speed squared before resolving the rebound angle.

Complete reasoning route

1. REPRESENT - Mark impact as a new launch event.
2. DECIDE - Use $v_{\text{after}}^2 = 0.60 v_{\text{before}}^2$.
3. TRANSFORM - Resolve post-bounce vertical speed and use $H = v_y^2 / (2g)$.
4. VERIFY - Height must be lower than the original and scale with $0.60 \sin^2(30 \text{ degrees})$.

Answer

Original 80 m at 45 degrees gives $u^2 = 320g$. After impact $v^2 = 192g$; $v_y^2 = 48g$; $H = 24 \text{ m}$.

Independent check

24 m is less than 80 m and has the expected energy/component scaling.

Practice 3

An ordinary projectile would have range R . At the apex it enters a region where downward gravity becomes $4g$ while horizontal velocity is unchanged. Express the new range as a fraction of R .

Your independent attempt

Crux move

Apply $4g$ only after the apex; do not alter stage 1.

Complete reasoning route

1. REPRESENT - Split the event at the apex.
2. DECIDE - Apply $4g$ only to the post-apex descent.
3. TRANSFORM - The descent time halves, so post-apex horizontal distance becomes $R/4$.
4. VERIFY - Total range must lie between $R/2$ and R .

Answer

New range = $R/2 + R/4 = 3R/4$.

Independent check

Only the descent segment changes, so $3R/4$ is physically bounded correctly.

Practice 4

A projectile launched at 45 degrees passes through $P=(4\text{ m}, 1\text{ m})$. Find u^2/g and decide whether the projectile is rising or falling at P.

Your independent attempt

Crux move

Point membership and branch classification are separate decisions.

Complete reasoning route

1. REPRESENT - Use the gravity-only trajectory equation.
2. TRANSFORM - At 45 degrees, $y = x - gx^2/u^2$; impose P.
3. DECIDE - Use local slope or v_y at P to identify the branch.
4. VERIFY - Substitute the result back into the path.

Answer

$1 = 4 - 16g/u^2$ gives $u^2/g = 16/3$ m. The slope is $1 - 2gx/u^2 = -1/2$, so the projectile is descending.

Independent check

Back-substitution gives $y=1$, and the negative slope places P after the apex.

Practice 5

A projectile has horizontal launch component 120 m/s. In a medium with linear drag, $m/c=2$ s. Find its horizontal displacement after 2 s using $e=2.718$.

Your independent attempt

Crux move

Use exponential velocity decay, not constant horizontal velocity.

Complete reasoning route

1. REPRESENT - Write $v_x = v_{x0} \exp(-t/\tau)$, $\tau = m/c = 2$ s.
2. DECIDE - Reject constant-acceleration/projectile $x = v_{x0} t$.
3. TRANSFORM - Integrate v_x from 0 to 2 s.
4. VERIFY - Distance must be less than the no-drag 240 m.

Answer

$\Delta x = 120(2)(1 - e^{-1}) = \text{about } 151.7 \text{ m, approximately } 152 \text{ m.}$

Independent check

The distance is below 240 m and approaches $v_{x0} t$ as drag tends to zero.